

Lorin Urbantat

Computer Science Student @ ETH Zurich · Software Engineer @ ETH-CES

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Education

ETH Zurich, MSc Computer Science - Expected Graduation: September 2027 - Major: Data Management Systems - Information Security Lab (5.75), Computational Semantics for Natural Language Processing (5.75), Advanced Formal Language Theory (5.5)	2025 - 2027
ETH Zurich, BSc Computer Science - Graduated August 2024, GPA 5.04 (6.0 – highest) - Parallel Programming (5.75), Numerical Methods (5.75), Analysis (5.5), Visual Computing (5.75), Applied Computer Architecture (6)	2021 - 2024
Waldorfschule Mülheim an der Ruhr, Abitur - GPA 1.3 (1.0 – highest)	2007 - 2020
Deutsche Schüler Akademie - Summer school for gifted students	2019

WORK EXPERIENCE

Information Security Lab Teaching Assistant, ETH Zurich - Test lab exercises and host exercise sessions	2026 - Present
Software Engineer, ETH Chair of Education Systems - Sole engineer responsible for end-to-end development of a research-focused RAG chatbot using Next.js, Convex and Microsoft Foundry, serving 1,000 users	2025 - Present
Software Engineer, ETH Game Technology Center - Morph Tales Game, Unity 3D - Developed Unity features for Morph Tales and built a game analytics pipeline using Firebase, BigQuery and Looker Studio.	2023 - 2025
Coach, mint & pepper - STEM and AI tutor for high school students	2025
Calculus Teaching Assistant and Exam-Preparation Course, ETH Zurich - Host exercise classes, 15 - 30 students	2023
Tutor, Waldorfschule Mülheim an der Ruhr - Tutorial classes (10 - 25 participants) in Math and Computer Science	2017 - 2019

PERSONAL PROJECTS & RELEVANT EXPERIENCE

Unikraft on Raspberry Pi 5, ETH Semester Project - Ported Unikraft to Raspberry Pi 5, implementing AArch64 boot, memory initialization, interrupts and timers - Developed a cache-safe kernel reload mechanism using OpenOCD and GDB, enabling repeated deployment without power-cycling.	2026
Flateroo, flat sharing platform - React Native application with real-time chat, video streaming infrastructure - Node backend, video transcoding, HLS streaming, PostgreSQL, fly.io	2024 - 2025
OpenDemocracy, RAG Powered Voting Advice application for elections	2024
SCION Control Plane Algorithms Test Environment - Built a Docker-based environment and traffic generator to evaluate SCION control-plane algorithms under configurable topologies and changing link conditions.	2024
TLDR – Lecture Summarizer (VIScon Hackathon) - Lecture summarizer using OpenAI Whisper to create searchable lecture summaries from videos	2023

SKILLS & INTERESTS

Languages	German (native), English (fluent), French (DELFI B1)
Technologies	AArch64 Assembly, C, C++, Kernel Development, TypeScript, React, Python, Java, Git, Linux, Containerization, Unity 3D, React Native, Node.js, SQL, RAG, Machine Learning, PCA
Interests	Biking, Sports, Cooking, Dancing, Singing